

CyberChef — Practical Tool Guide

Base64, Hex, URL/HTML decoding, Magic, XOR, hashes, and recipes as a fast browser-based analysis and transformation reference.

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Usage note: CyberChef is useful for analyzing and transforming data in the browser; it does not require commands to be run against a target.

INPUT

Data

OPERATION

Choose transform

RECIPE

Order steps

OUTPUT

Validate

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1. CyberChef interface

Think in four areas: Input, Operations, Recipe, and Output.

Area	What you do
Input	Paste the text/data.
Operations	Find operations such as From Base64, From Hex, URL Decode, XOR, or Magic.
Recipe	Order operations in sequence.
Output	Inspect the result immediately.

CyberChef is excellent for decoding and transformations, but it does not “know” the intended answer. Validate the result against the lab context.

2. Core Encoding / Decoding

Type	Operation	Common use
Base64	To Base64 / From Base64	Cookies, tokens, data.
Hex	To Hex / From Hex	Bytes and hexdumps.
URL	URL Encode / URL Decode	Path, query, and parameters.
HTML	From HTML Entity	Entities such as ".
UTF-8 / text	Character encoding operations	When text appears corrupted or mis-decoded.

3. Magic

When the encoding is unknown, Magic is a good starting helper, then validate manually.

CYBERCHEF — Recipe

Magic

Magic can suggest operations for data that resembles Base64, Hex, Gzip, and other encodings. A suggestion is not a security finding by itself.

4. XOR and Hashes

Use these for analysis/comparison and do not confuse a hash with reversible encoding.

CYBERCHEF — Hash a value

MD5 / SHA1 / SHA2

Choose the required algorithm and compare the digest.

CYBERCHEF — XOR

XOR

Set the correct key and key format based on the challenge context.

Note: Hashes are generally one-way; Base64/Hex are encodings. Do not try to “decode SHA256” as if it were Base64.

5. Useful recipes

The sequence of operations is where CyberChef becomes especially useful.

CYBERCHEF — Nested encoding example

From Base64
From Hex
URL Decode

This is only an example order; apply it when the data format justifies it.

CYBERCHEF — Multiple lines

Fork → From Base64 → Merge

Fork is useful when many values need the same recipe.

6. Workflow

Step	Action	Note
1	Paste	Keep a copy of the original value.
2	Recognize	Does it look like Base64, Hex, or URL encoding?
3	Magic	Use it when the format is unclear.
4	Recipe	Apply one operation and observe the result.
5	Validate	Does the output make sense in context?
6	Record	Save the important recipe/steps.

Prepared as a quick-reference companion to the canonical Arabic guide. Use only in labs or systems you are explicitly authorized to test.

References:

<https://github.com/gchq/CyberChef>

<https://github.com/gchq/CyberChef/wiki/Automatic-detection-of-encoded-data>